BIOMECHANICAL ANALYSIS OF THE STANDING LONG JUMP

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ABSTRACT

The purposes of the present study were to (1) investigate the effects of the arm movement and initial knee joint angle employed in standing long jump by the ground reaction force analysis and three-dimensional motion analysis; and (2) investigate how the jump performance of the female gender related to the body configuration. Thirty-four healthy adult females performed standing long jump on a force platform with full effort. Body segment and joint angles were analyzed by three-dimensional motion analysis system. Using kinetic and kinematic data, the trajectories on mass center of body, knee joint angle, magnitude of peak takeoff force, and impulse generation in preparing phase were calculated. Average standing long jump performances with free arm motion were +1.5 times above performance with restricted arm motion in both knee initial angles. The performances with knee 90° initial flexion were +1.2 times above performance with knee 45° initial flexion in free and restricted arm motions. Judging by trajectories of the center mass of body (COM), free arm motion improves jump distance by anterior displacement of the COM in starting position. The takeoff velocity with 90° knee initial angle was as much as 11% higher than in with 45° knee initial angle. However, the takeoff angles on the COM trajectory showed no significant differences between each other. It was found that starting jump from 90° bend knee relatively extended the time that the force is applied by the leg muscles. To compare the body configurations and the jumping scores, there were no significant correlations between jump scores and anthropometry data. The greater muscle mass or longer leg did not correlated well with the superior jumping performance.

Biomed Eng Appl Basis Comm, 2003 (October); 15: 186-192.
Keywords: Standing long jump, Kinematics, Ground reaction force, Biomechanics

1. INTRODUCTION

The performance of standing long jump was often adopted to examine the fitness of the school children [1-2], but the test frequently underestimate the subject's true potential when the subject did not use the best possible technique. The selection of optimum takeoff angle and technical use of the arms are two of the most important methods that promote the jumping performance. Previously, many researches used the force plates to study the long jump [3-4] and suggested the technique of long jump [5-8]. They concluded that increasing the takeoff velocity of the jumper's supporting leg would increase jumping distance. Other researchers used numerical model and found there

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were no significant differences among jumps with different inclination angles in the magnitude of the peak ground reaction force [9].

Previous standing long jump study compared body configuration and joint moment analysis in children and adult males [10]. They suggested that skilled 6-yr-old children have not developed either hip negative work during preparatory movement or body configuration in the flight phase. The difference in body configuration showed the jumping performance and cross-sectional of the skeletal muscle correlate positively [4]. Researchers also verified that standing long jump performance improved when arm movement was employed [11].

Many recent studies investigated the explosive movements concerned mainly with vertical jump. Although jumping for distance received a lot of attention, most of these works were center on long or triple jump in athletics. As far as we know there were no attempt to relate the coordination of hands swing and starting knee angles in standing jumping distance. Also the difference in initial knee angle in this study was not been investigated in other studies. The performance of standing long jump and its relationship to the body configurations as well as the possible other reasons for the differences are still not so clear. Furthermore, there is no information about trajectory of body center of mass in the standing long jump, but other basic motor skills (running, walking, kicking, and vertical jump) had been investigated by using the center of mass analysis. Therefore, the study efforts were directed towards to (1) investigating the effects on the arm movement and initial knee joint angle employed in standing long jump by the ground reaction force analysis, three-dimensional motion analysis, and analyzing changes in force-time characteristics, magnitude of peak ground reaction forces, impulses generated in preparing phase and mass center's pathway characteristics during standing long jump; (2) investigating how the jump performance related to the body configuration.

2. METHODOLOGY

2.1 Subjects

Thirty-four unskilled healthy female collegiate students (mean age, 19.7±0.96 years) participated in this study. Their mean height and body weight were 160±5.99 cm, 50.71±10.83 kg (Table 1). The subjects were informed of the experimental procedures and gave their consent before experiment.

2.2 Protocol

Anthropometrical data was recorded for normalization of force data with respect to body weight. The shank length was defined as the distance from the fibular head to lateral malleous. The leg length was defined as the distance from the anterior superior iliac spine (ASIS) to the medial malleous. The shank and thigh circumferences were assessed through the widest areas of the shank and thigh. For kinematic analysis, twenty-one reflective landmarks (Helen Hayes) were attached to the trunk, upper limbs, and lower limbs. The jump motions were filmed (60Hz, 3 seconds) using a high-speed motion analysis system (ExpertVisionTM Hi-Res motion analysis system). Subjects performed maximal-effort barefoot standing long jumps three times every trial on a force platform (Kistler Instrument Corp., Switzerland). The subject stands with her toes immediately behind the take-off line at the starting position. Subjects were asked to choose their initial squatted position in 45° knee flexion or 90° knee flexion to see whether it was altered in relation to the score of jump. By bending the knees with free swinging the arms one time or with restricted arms, the subject jumps forward to cover the greatest distance possible. The distance is measured from the take-off line to the rearmost heel. The subject must complete three correctly executed jumps for each condition: subjects restricted their arms on their back with initial bending angles of the knee joint of 45°(45RA) and 90° (90RA), and two conditions with instruction to make arms free motion with initial

Table 1: The mean age and anthropometric data of thirty-four healthy female collegiate students participated in this study.

	Age -years	Height -cm	Shank length -cm	Thigh circumfrence -cm	Shank ircumference -cm	Leg Length -cm	Body Weight -kg
Mean	19.7	160	35.3	47.3	30.1	82.35	50.71
SD	0.96	5.99	3.24	5.31	3.08	4.49	10.83

bending angles of the knee joint of 45° (45FA) and 90° (90FA). The order of the four conditions was randomized to minimize effects of learning and fatigue. The knee angles were estimated by the naked eye. The deviations were estimated by the three-dimensional motion analysis later. The subjects were allowed three practice jumps in each condition. Subsequently, three jump scores were registered per condition. The subject must take off with both feet simultaneously and land on both feet simultaneously with the heels approximately even. If the subject lost her balance and falls backward, another trial was permitted.

2.3 Analysis

Jump scores

The records in the three trials were averaged.

Kinematic analysis

The position data of all markers were smoothed using a generalized cross-validation spline smoothing (GCVSPL) routine at a cutoff frequency of 6 Hz [12-13]. We registered the positions of the eleven body segments: feet, lower legs, upper legs, lower arms, upper arms and head-trunk in three-dimensional coordinates. Further analysis was limited to the sagittal plane, thus taking left and right as one. Joint angles of the knee were calculated using the stored coordinates. The position of whole body center of mass (COM) and the first time derivate of position data of COM were calculated by self-developed program based on body segments' parameters provided by Miller and Nelson [14]. Takeoff velocities of COM were obtained by calculating the velocity in three diagonal directions just before takeoff phase. To determine the takeoff angle, the instant run forward (v) and jump upward (u) velocities were measured. Putting these values in the equation ($\tan q = u/v$) for tan q we obtain a value for q.

Ground reaction force analysis

Vertical and horizontal (jumping direction) force signals were digitized and normalized by the body weight for later analysis.

Impulse

The vertical and horizontal impulse data for the jump were calculated by self-developed program. They are the force-time integral values from the onset of preparatory movement to the point of takeoff. The force multiplied by 1/1000 (as the force plates recorded the data at 1000 frames per second) and all summed to calculate the total impulse for the vertical and horizontal impulses. Vertical force contributed to the vertical impulse and horizontal force contributed to the horizontal impulse. Vertical impulse was calculated after subtracting the body weight from the measured vertical force. It represented by the total area over the body weight line (the vertical force in the standing phase represented by a horizontal line on the graph). The vertical force was out of consideration if it was lower than the body weight.

Statistics

Pearson correlation analysis was used to correlate between values. Difference between mean values was tested by repeated two-way ANOVA.

3. RESULTS

3.1 Body Configuration During Jumping

See Figure 1 for description of body center of mass relative to the body configuration at the instant of starting position, preparing phase, push-off phase, toe-off phase and touch-down. In this figure, four activities were obtained by using the trails performed by one person. Knee joint angles at the preparing phase were 43.18+3.03 degrees, 89.22+4.66 degrees in the trials with knee 45° flexion and 90° flexion, respectively. The difference in joint angles was not obvious between actual value and estimated value.

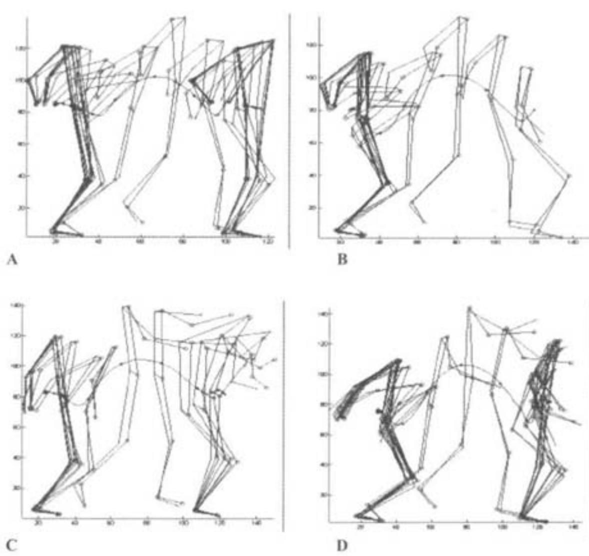

Figure 1: Body configuration from initial bend knee position to touch down. In these figures, four activities were obtained by using the trails performed by one person. (A) (top left panel), description of the kinematics of the mass center of the body in jumping with restricted arm motion and bending angles of the knee joint of 45° and (B) 90° (top right panel), and two conditions with instruction to make a free arm motion with (C) bending angles of the knee joint of 45° (bottom left panel) and (D) 90° (bottom right panel).

3.2 Ground Reaction Force During Jump

The effect of gravity on the direction of the resultant force is illustrated in Fig. 2. It displays the decomposition of the ground reaction force in three components. During push-off phase, there were peak force values in both the vertical and horizontal (anterior/posterior) directions, as well as value of resultant force. In the vertical and horizontal force-time curves, we found there were significant differences between 90° knee initial angle and 45° knee initial angle in the time duration that the force is applied in the preparing phase. The time period of preparing phase in 45° knee initial angle was lower ($p<0.05$) than in 90° knee initial angle. In the vertical force, the peak force during push-off phase in 90° knee initial angle was lower ($p<0.05$) than in 45° knee initial angle. Peak values of vertical ground reaction force relative to body weight were 2.21 ± 029 body weight^{-1} and 2.09 ± 0.26 body weight^{-1} in the trials of knee 45°flexion with restricted arm motion and knee 45° flexion with free arm motion (Figure 3). They were 1.50 ± 0.26 body weight^{-1} and 1.62 ± 0.30 body weight^{-1} in the trials of knee 90° flexion with restricted arm motion and knee 90° flexion with free arm motion. There were significant differences in the magnitude of peak vertical force between 90° and 45° knee flexion initial angle ($p<0.05$), but no significant difference between arms motions ($p>0.05$). In the magnitude of peak horizontal force, there was no significant difference ($p>0.05$) among jumps with each technique (Figure 4).

3.3 Impulse

The impulses generated in preparing phase are shown in Figure 5 and 6. The results showed that vertical and horizontal impulses were higher in 90° knee flexion jumping than 45° knee flexion jumping ($p<0.01$).

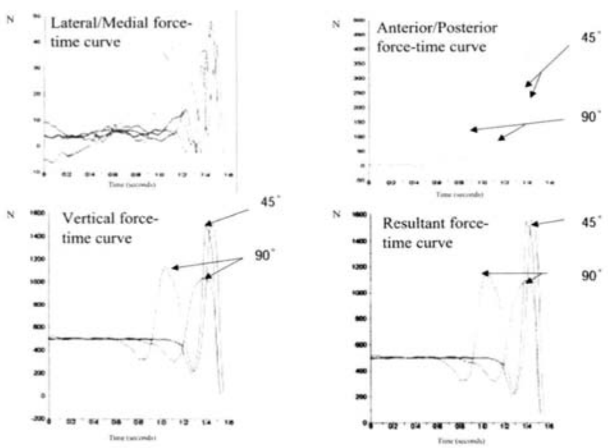

Figure 2: Typical force-time history for a standing long jump. The curves of three orthogonal force components were studied.

	Restricted Arms		Free Arms	
	45 degrees	90 degrees	45 degrees	90 degrees
Mean	2.21	1.50	2.09	1.62
SD	0.29	0.26	0.26	0.30

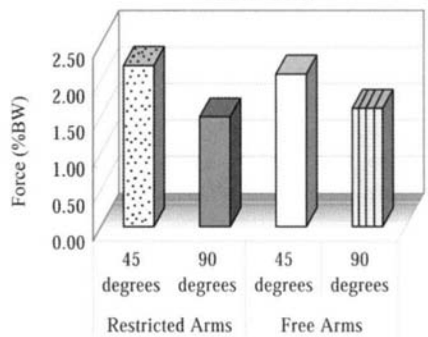

Figure 3: Peak values of vertical ground reaction force relative to body weight.

	Restricted Arms		Free Arms	
	45 degrees	45 degrees	90 degrees	90 degrees
Mean	0.63	0.66	0.65	0.70
SD	0.15	0.10	0.11	0.11

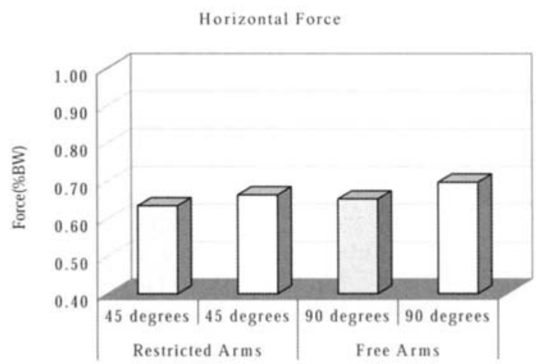

Figure 4: Peak values of horizontal ground reaction force relative to body weight.

3.4 Jump Scores

Average jump records of knee 45° flexion with free arm motion, knee 45° flexion with restricted arm motion, knee 90° flexion with free arm motion, and knee 90° flexion with restricted arm motion were 68.23 ± 20.12 cm, 51.70 ± 12.79 cm, 93.35 ± 20.38 cm, and 80.25 ± 15.38 cm respectively (Figure 7). The distances in standing long jump were ranked as follows: 90° knee initial angle with free arm motion were longer ($P < 0.01$) than in 90° knee initial angle with restricted arm motion; subsequently were 45° knee initial angle with free arm motion; and the last was 45°

	Restricted Arms		Free Arms	
	45 degree	90 degree	45 degree	90 degrees
Mean	82.39	98.13	86.72	100.17
SD	22.58	22.18	18.55	26.04

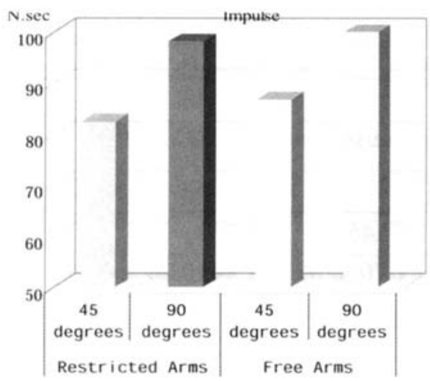

Figure 5: Vertical impulse generated in preparing phase. It was higher in 90° knee flexion jumping than 45° knee flexion jumping.

	Restricted Arms		Free Arms	
	45 degree	90 degree	45 degree	90 degrees
Mean	73.23	84.26	73.54	86.29
SD	15.33	17.27	16.02	13.66

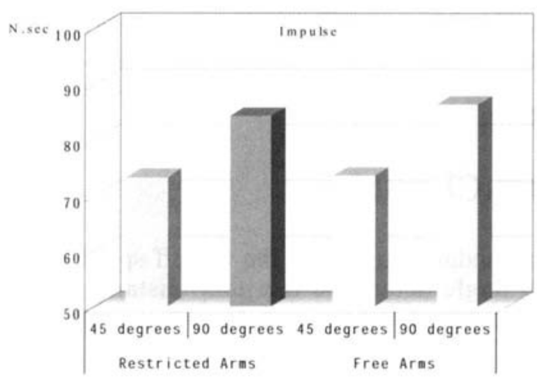

Figure 6: Horizontal impulse generated in preparing phase. It was higher in 90° knee flexion jumping than 45° knee flexion jumping.

knee initial angle with restricted arm motion ($P < 0.05$). Average standing long jump performances with free arm motion were +1.5 times above performance with restricted arm motion in both knee initial angles. The performances with knee 90° initial flexion were +1.2 times above performance with knee 45° initial flexion in with free and restricted arm motions.

	Restricted Arms		Free Arms	
	45 degrees	90 degrees	45 degrees	90 degrees
Mean (cm)	51.70	80.25	68.23	93.35
SD	12.79	15.38	20.12	20.38

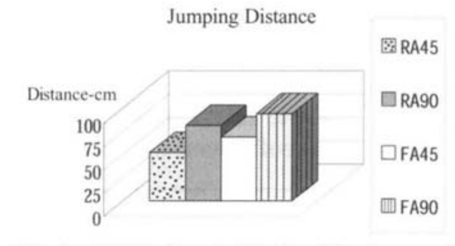

Figure 7: Average jump records of knee 45° flexion with free arm motion, knee 45° flexion with restricted arm motion, knee 90° flexion with free arm motion, and knee 90° flexion with restricted arm motion.

3.5 Anthropometric Parameters vs. Score

The mean dimensions of subject limbs respectively were 35.3 cm for shank length, 47.3 cm for thigh circumference, 30.1 cm for shank circumference, 82.35 cm for leg length, and 50.71 kg for body weight (Table I). There were no significant correlations between jump scores and anthropometry (Table II, $r<0.49$). The greater muscle mass or longer leg was not the sole factor that affect the superior jumping performance.

3.6 Takeoff Angle and Velocity During Jumping

The trajectories of the COM are shown in Figure 8. The average takeoff angle and velocity are illustrated in Table III. The rate of force development (takeoff velocity) on the COM trajectory was in with 45° knee initial angle lower ($P < 0.01$) than in with 90° knee initial angle. When the performances of 45° and 90° knee initial angle were grouped by with free or restricted arm motions, it was found that the takeoff velocity on the COM trajectory with 90° knee initial angle was 11% higher than that of with 45° knee initial angle. However, the takeoff angles on the COM trajectory showed no significant differences among each technique.

4. DISCUSSION

4.1 Performance in the Standing Long Jump

The performance in the standing long jump is evaluated by the total jump distance, which is the

Table II: The dependence (correlation coefficients) of the jumping distance and anthropometric data.

	Height	Shank length	Thigh Circumference	Shank Circumference	Leg Length	Body Weight
RA45	0.25	0.27	0.39	0.28	0.36	0.49
RA90	0.18	0.41	0.37	0.28	0.17	0.32
FA45	0.29	0.31	0.26	0.17	0.21	0.19
FA90	0.24	0.4	0.25	0.15	0.12	0.16

Table III: The average magnitude of the velocity (m/sec) of the COM at takeoff phase and the takeoff angle (degrees).

	RA45	RA90	FA45	FA90
Takeoff velocity (m/sec)	1.64±0.11	1.78±0.18	1.70±0.07	1.94±0.24
Take off angle (degrees)	48.00±2.00	50.67±2.52	49.00±1.00	49.33±3.06

horizontal distance from takeoff line to the mark made by the heel at landing. It is the sum of three component distances: takeoff distance, flight distance, and landing distance. The horizontal distance from the takeoff toe to the takeoff COM position made up the takeoff distance. The horizontal distance from landing COM position to the landing heel makes up the landing distance. In standing long jumps with restricted arm motion, the jumping distances decreased (Figure 7) because of lower magnitude of horizontal component of the jumper's COM vector, which was estimated from the toe marker to the COM position. It was one of the reason why flight distance in standing long jump with 45° knee initial flexion angle results in the lower jumping scores.

4.2 Impulse vs. Score

If the long jumper wants to jump as far as possible, it would be best for her to apply as much as force as possible at takeoff over the longest available time frame. The greatest difference between the four testing conditions was observed between the timing of the various jumping movements (Figure 2). The better performers provided increase initial impulses, but the poorer performers, the impulse decreased during preparing phase (Figure 5, 6). Starting jump from 90° bend knee would increase the time that the force was applied by the leg muscles. As a result that the vertical and horizontal impulses were higher in 90° knee flexion jumping than 45° knee flexion jumping (Figure 5, 6), this explains why the jumpers perform better in 90° initial angle than in 45° knee initial angle (Figure 7).

4.3 Swing of the Upper Extremity vs. Score

Preparation move before jump started with backward swing of the arms that resulted in the shift of the COM anterior to the edge of the supporting base (Figure 8). Ashby et al. [11] reported pre-takeoff increases of 8 cm in the horizontal displacement of the center of gravity. The present study showed an average increase of 7.5 cm in the horizontal displacement of the center of mass before takeoff when arm motion was allowed. This highly unstable position satisfies two pre-requirements. It extended jumper's legs into a powerful thrusting stance. Second, it put the jumper into a position that requires minimal force to thrust him/her in the required direction, toward the greatest distance. Forward swing of the arms before jumping, jumper generates an angular momentum that was then transferred to his/her body as a whole. The forward swing of the arms adds to the thrust of the leg during push down.

5. CONCLUSION

In standing long jump, the takeoff speed and the takeoff angle determine the jump distance. In this study we found the takeoff velocity on the COM

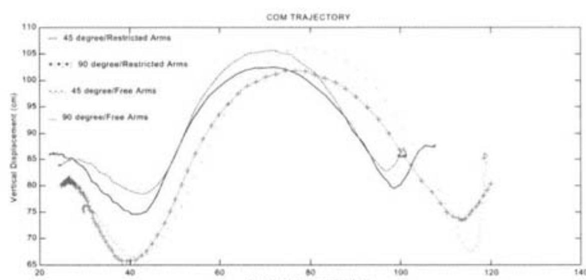

Figure 8: The trajectory s of the mass center of the body (COM).

trajectory was lower in 45° initial knee angles than in 90° initial knee angles. The takeoff angles on the COM trajectory showed no significant differences between each other. The results showed that jumping with 90° knee initial angle with free arm motion creates the longest distances in standing long jump. This was followed by jumping with 90° initial knee angle with restricted arm motion; 45° initial knee angle with free arm motion; and lastly 45° initial knee angle with restricted arm motion.

Starting the jump from 90° bend knee increased the time that the force was applied by the leg muscles. As a result the vertical and horizontal impulse was higher in 90° flexion knee than 45° knee flexion jumping. This explained why the jumper performed better when the knee bend 90° in pre-jump stage than 45° initial knee angle.

To compare the body configurations and the jumping scores in adult females, there were no significant correlations between jump scores and anthromotry parameters. The greater muscle mass or longer lower leg did not correlate well with the superior jumping performance.

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